

atividade_aula11

```
# Importando os dados
```

```
library(readxl)
```

```
dados <- read_excel("dados_aula_campo.xlsx", skip = 2)
```

```
## New names:
```

```
## * `m³/m³ Water Content` -> `m³/m³ Water Content...16`
```

```
## * `°C Soil Temperature` -> `°C Soil Temperature...17`
```

```
## * `mS/cm Bulk EC` -> `mS/cm Bulk EC...18`
```

```
## * `m³/m³ Water Content` -> `m³/m³ Water Content...19`
```

```
## * `°C Soil Temperature` -> `°C Soil Temperature...20`
```

```
## * `mS/cm Bulk EC` -> `mS/cm Bulk EC...21`
```

```
## * `m³/m³ Water Content` -> `m³/m³ Water Content...22`
```

```
## * `°C Soil Temperature` -> `°C Soil Temperature...23`
```

```
## * `mS/cm Bulk EC` -> `mS/cm Bulk EC...24`
```

```
## * `°C Soil Temperature` -> `°C Soil Temperature...26`
```

```
dados
```

```
## # A tibble: 7,413 x 31
```

```
##   Timestamp          `W/m² Solar Radiation` `mm Precipitation`
```

```
##   <dtm>                <dbl>                <dbl>
```

```
## 1 2025-11-20 15:00:00          670.                0
```

```
## 2 2025-11-20 16:00:00          478.                0
```

```
## 3 2025-11-20 17:00:00          148.                0
```

```
## 4 2025-11-20 18:00:00           52.8                0
```

```
## 5 2025-11-20 19:00:00            1                0
```

```
## 6 2025-11-20 20:00:00            0                0
```

```
## 7 2025-11-20 21:00:00            0                0
```

```
## 8 2025-11-20 22:00:00            0                0
```

```
## 9 2025-11-20 23:00:00            0                0
```

```
## 10 2025-11-21 00:00:00           0                0
```

```
## # i 7,403 more rows
```

```
## # i 28 more variables: `Lightning Activity` <dbl>,
```

```
## #   `km Lightning Distance` <dbl>, `° Wind Direction` <dbl>,
```

```
## #   `m/s Wind Speed` <dbl>, `m/s Gust Speed` <dbl>, `°C Air Temperature` <dbl>,
```

```
## #   `RH Relative Humidity` <dbl>, `kPa Atmospheric Pressure` <dbl>,
```

```
## #   `° X-axis Level` <dbl>, `° Y-axis Level` <dbl>,
```

```
## #   `mm/h Max Precip Rate` <dbl>, `°C RH Sensor Temp` <dbl>, ...
```

```
# Verificando a estrutura do arquivo
```

```
str(dados)
```

```
## tibble [7,413 x 31] (S3: tbl_df/tbl/data.frame)
```

```
## $ Timestamp          : POSIXct[1:7413], format: "2025-11-20 15:00:00" "2025-11-20 16:00:00" ..
```

```
## $ W/m² Solar Radiation : num [1:7413] 669.5 478.4 147.8 52.8 1 ...
```

```
## $ mm Precipitation    : num [1:7413] 0 0 0 0 0 0 0 0 0 ...
```

```
## $ Lightning Activity      : num [1:7413] 0 0 0 0 0 0 0 0 0 0 ...
## $ km Lightning Distance  : num [1:7413] 0 0 0 0 0 0 0 0 0 0 ...
## $ ° Wind Direction       : num [1:7413] 273 271 272 270 272 271 269 270 271 271 ...
## $ m/s Wind Speed         : num [1:7413] 9.82 9.7 8.57 7.37 5.75 4.84 4.49 4.24 4.3 4.04 ...
## $ m/s Gust Speed         : num [1:7413] 11.46 11.28 10.95 8.26 6.77 ...
## $ °C Air Temperature     : num [1:7413] 37 37 35.8 33.2 28.2 ...
## $ RH Relative Humidity   : num [1:7413] 0.292 0.299 0.306 0.359 0.475 ...
## $ kPa Atmospheric Pressure: num [1:7413] 99.7 99.6 99.5 99.5 99.5 ...
## $ ° X-axis Level         : num [1:7413] 1.8 1.7 1.7 1.8 1.8 1.9 1.8 1.8 1.8 1.8 ...
## $ ° Y-axis Level         : num [1:7413] -1.7 -1.6 -1.6 -1.5 -1.5 -1.5 -1.5 -1.5 -1.5 -1.5 ...
## $ mm/h Max Precip Rate   : num [1:7413] 0 0 0 0 0 0 0 0 0 0 ...
## $ °C RH Sensor Temp      : num [1:7413] 39.4 39.5 37.6 34.6 29.2 ...
## $ m³/m³ Water Content...16: num [1:7413] 0.457 0.453 0.45 0.448 0.447 ...
## $ °C Soil Temperature...17: num [1:7413] 27.1 27.1 27.1 27.1 27.1 27.1 27.1 27.1 27.1 27.1 ...
## $ mS/cm Bulk EC...18     : num [1:7413] 0.942 0.92 0.914 0.91 0.912 ...
## $ m³/m³ Water Content...19: num [1:7413] 0.453 0.453 0.453 0.453 0.453 ...
## $ °C Soil Temperature...20: num [1:7413] 26.8 26.9 27 27.1 27.3 ...
## $ mS/cm Bulk EC...21     : num [1:7413] 0.726 0.726 0.726 0.726 0.728 ...
## $ m³/m³ Water Content...22: num [1:7413] 0.317 0.317 0.317 0.316 0.316 ...
## $ °C Soil Temperature...23: num [1:7413] 29.5 30.2 30.4 30.3 30 ...
## $ mS/cm Bulk EC...24     : num [1:7413] 0.545 0.545 0.543 0.54 0.538 ...
## $ kPa Matric Potential    : logi [1:7413] NA NA NA NA NA NA ...
## $ °C Soil Temperature...26: logi [1:7413] NA NA NA NA NA NA ...
## $ Sensor Output          : num [1:7413] 0 0 0 0 0 0 0 0 0 0 ...
## $ % Battery Percent       : num [1:7413] 100 100 100 100 100 100 100 100 100 100 ...
## $ mV Battery Voltage      : num [1:7413] 8044 8028 8010 7993 7979 ...
## $ kPa Reference Pressure  : num [1:7413] 99.6 99.5 99.4 99.4 99.5 ...
## $ °C Logger Temperature   : num [1:7413] 41.7 41.9 38.5 35.3 29.2 ...
```

```
# Verificando dados ausentes
```

```
all(is.na(dados))
```

```
## [1] FALSE
```

```
# Selecionando as variáveis de interesse
```

```
dados_campo <- dados[, c("Timestamp", "°C Air Temperature", "mm Precipitation")]
```

```
# Renomeando as variáveis
```

```
names(dados_campo) <- c("horario", "temperatura_ar", "precipitacao")
```

```
# Exibindo a estrutura do novo dataframe
```

```
str(dados_campo)
```

```
## tibble [7,413 x 3] (S3: tbl_df/tbl/data.frame)
```

```
## $ horario      : POSIXct[1:7413], format: "2025-11-20 15:00:00" "2025-11-20 16:00:00" ...
```

```
## $ temperatura_ar: num [1:7413] 37 37 35.8 33.2 28.2 ...
```

```
## $ precipitacao  : num [1:7413] 0 0 0 0 0 0 0 0 0 0 ...
```

```
# Selecionando o período de interesse
```

```
data1 <- "2026-01-01 00:00:00"
```

```
data2 <- "2026-09-25 00:00:00"
```

```
dados_perodo_interesse <- dados_campo[
```

```

dados_campo$horario >= as.POSIXct(data1)
& dados_campo$horario <= as.POSIXct(data2),
]

# Vendo se o corte foi feito corretamente
head(dados_perodo_interesse)

## # A tibble: 6 x 3
##   horario                temperatura_ar precipitacao
##   <dtm>                  <dbl>          <dbl>
## 1 2026-01-01 04:00:00      24.8            0
## 2 2026-01-01 05:00:00      25.0            0
## 3 2026-01-01 06:00:00      25.0            0
## 4 2026-01-01 07:00:00      26.2            0
## 5 2026-01-01 08:00:00      29.3            0
## 6 2026-01-01 09:00:00      31.2            0

tail(dados_perodo_interesse)

## # A tibble: 6 x 3
##   horario                temperatura_ar precipitacao
##   <dtm>                  <dbl>          <dbl>
## 1 2026-09-24 23:00:00      23.8            0
## 2 2026-09-25 00:00:00      22.5            0
## 3 2026-09-25 01:00:00      21.2            0
## 4 2026-09-25 02:00:00      20.2            0
## 5 2026-09-25 03:00:00      19.6            0
## 6 2026-09-25 04:00:00      19.2            0

# Calculando as temperaturas médias, mínimas e máximas diárias

library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

temperaturas_medias_diarias <- dados_perodo_interesse %>%
  mutate(data = as.Date(horario)) %>%
  group_by(data) %>%
  summarise(
    temperatura_media = mean(temperatura_ar, na.rm = TRUE),
    .groups = "drop"
  )

temperaturas_minimas_diarias <- dados_perodo_interesse %>%
  mutate(data = as.Date(horario)) %>%
  group_by(data) %>%
  summarise(
    temperatura_minima = min(temperatura_ar, na.rm = TRUE),

```

```

    .groups = "drop"
  )

temperaturas_maximas_diarias <- dados_perodo_interesse %>%
  mutate(data = as.Date(horario)) %>%
  group_by(data) %>%
  summarise(
    temperatura_maxima = max(temperatura_ar, na.rm = TRUE),
    .groups = "drop"
  )

```

```
temperaturas_medias_diarias
```

```

## # A tibble: 268 x 2
##   data      temperatura_media
##   <date>          <dbl>
## 1 2026-01-01      30.1
## 2 2026-01-02      27.5
## 3 2026-01-03      28.3
## 4 2026-01-04      28.6
## 5 2026-01-05      25.1
## 6 2026-01-06      26.5
## 7 2026-01-07      29.0
## 8 2026-01-08      29.4
## 9 2026-01-09      28.0
## 10 2026-01-10     27.7
## # i 258 more rows

```

```
temperaturas_minimas_diarias
```

```

## # A tibble: 268 x 2
##   data      temperatura_minima
##   <date>          <dbl>
## 1 2026-01-01      24.8
## 2 2026-01-02      22.3
## 3 2026-01-03      22.8
## 4 2026-01-04      23.1
## 5 2026-01-05      16.8
## 6 2026-01-06      16.5
## 7 2026-01-07      21.9
## 8 2026-01-08      23.5
## 9 2026-01-09      23.3
## 10 2026-01-10     23.3
## # i 258 more rows

```

```
temperaturas_maximas_diarias
```

```

## # A tibble: 268 x 2
##   data      temperatura_maxima
##   <date>          <dbl>
## 1 2026-01-01      35.9
## 2 2026-01-02      36.6
## 3 2026-01-03      36.4
## 4 2026-01-04      35.9
## 5 2026-01-05      34.1

```

```
## 6 2026-01-06          36.8
## 7 2026-01-07          38.6
## 8 2026-01-08          37.0
## 9 2026-01-09          34.2
## 10 2026-01-10         36.9
## # i 258 more rows

# Calculando as estatísticas de temperatura mensais

library(lubridate)

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##      date, intersect, setdiff, union

temperaturas_mensais <- temperaturas_medias_diarias %>%
  left_join(temperaturas_maximas_diarias, by = "data") %>%
  left_join(temperaturas_minimas_diarias, by = "data") %>%
  mutate(
    ano = year(data),
    mes = month(data)
  ) %>%
  group_by(ano, mes) %>%
  summarise(
    temperatura_media_mensal =
      mean(temperatura_media, na.rm = TRUE),

    media_temperaturas_maximas_diarias =
      mean(temperatura_maxima, na.rm = TRUE),

    media_temperaturas_minimas_diarias =
      mean(temperatura_minima, na.rm = TRUE),

    .groups = "drop"
  )

knitr::kable(
  temperaturas_mensais,
  digits = 2,
  col.names = c(
    "Ano",
    "Mês",
    "Temperatura média (°C)",
    "Média das máximas diárias (°C)",
    "Média das mínimas diárias (°C)"
  )
)
```

		Temperatura média		
Ano	Mês	(°C)	Média das máximas diárias (°C)	Média das mínimas diárias (°C)
2026	1	27.98	35.39	22.23
2026	2	27.92	35.12	23.12

Ano	Mês	Temperatura média		
		(°C)	Média das máximas diárias (°C)	Média das mínimas diárias (°C)
2026	3	27.73	34.55	23.15
2026	4	26.52	33.21	21.83
2026	5	21.48	27.61	16.93
2026	6	20.64	27.42	15.56
2026	7	23.19	30.49	17.52
2026	8	24.76	33.23	17.54
2026	9	24.49	31.22	18.75

```
# Calculando a precipitação acumulada por dia

precipitacao_acumulada_diaria <- dados_perodo_interesse %>%
  mutate(data = as.Date(horario)) %>%
  group_by(data) %>%
  summarise(
    precipitacao_diaria = sum(precipitacao, na.rm = TRUE),
    .groups = "drop"
  )

precipitacao_acumulada_diaria
```

```
## # A tibble: 268 x 2
##   data      precipitacao_diaria
##   <date>          <dbl>
## 1 2026-01-01          0.170
## 2 2026-01-02          9.59
## 3 2026-01-03          1.04
## 4 2026-01-04           0
## 5 2026-01-05           0
## 6 2026-01-06         0.0170
## 7 2026-01-07           0
## 8 2026-01-08           0
## 9 2026-01-09         0.817
## 10 2026-01-10        22.4
## # i 258 more rows
```

```
# Calculando a precipitação acumulada mensal

precipitacoes_mensais <- precipitacao_acumulada_diaria %>%
  mutate(
    ano = year(data),
    mes = month(data)
  ) %>%
  group_by(ano, mes) %>%
  summarise(
    precipitacao_mensal =
      sum(precipitacao_diaria, na.rm = TRUE),

    .groups = "drop"
  )

knitr::kable(
```

```
precipitacoes_mensais,
digits = 2,
col.names = c(
  "Ano",
  "Mês",
  "Precipitação mensal (mm)"
)
)
```

Ano	Mês	Precipitação mensal (mm)
2026	1	121.60
2026	2	165.53
2026	3	109.38
2026	4	80.31
2026	5	51.63
2026	6	60.54
2026	7	15.48
2026	8	23.35
2026	9	35.07

```
# Calculando a precipitação total acumulada do período
```

```
sum(precipitacoes_mensais$precipitacao_mensal)
```

```
## [1] 662.8851
```

```
# Criando o gráfico de temperatura
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
temperaturas_mensais %>%
```

```
  pivot_longer(
```

```
    cols = c(
```

```
      temperatura_media_mensal,
```

```
      media_temperaturas_maximas_diarias,
```

```
      media_temperaturas_minimas_diarias
```

```
    ),
```

```
    names_to = "tipo_temperatura",
```

```
    values_to = "temperatura"
```

```
  ) %>%
```

```
  mutate(
```

```
    mes = factor(
```

```
      mes,
```

```
      levels = 1:12,
```

```
      labels = c(
```

```
        "Jan", "Fev", "Mar", "Abr", "Mai", "Jun",
```

```
        "Jul", "Ago", "Set", "Out", "Nov", "Dez"
```

```
      )
```

```
    )
```

```
  ) %>%
```

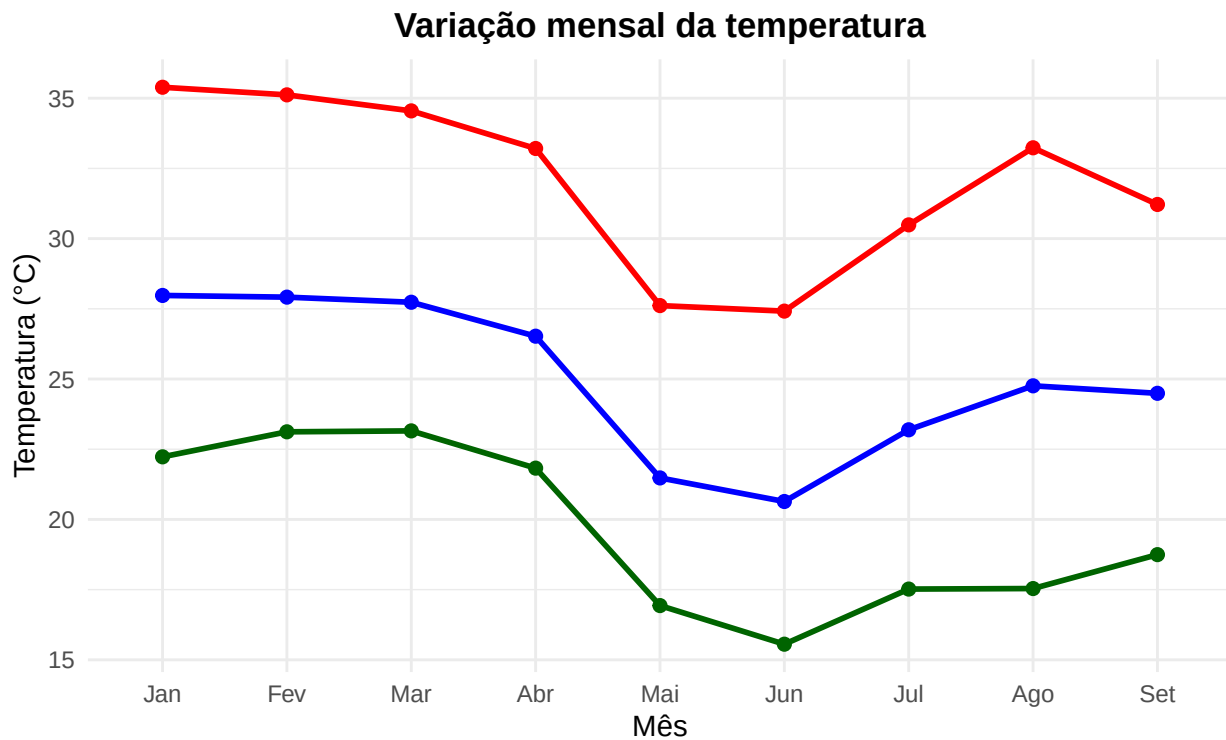
```
  ggplot(
```

```
    aes(
```

```

    x = mes,
    y = temperatura,
    color = tipo_temperatura,
    group = interaction(tipo_temperatura, ano)
  )
) +
geom_line(linewidth = 1) +
geom_point(size = 2) +
scale_color_manual(
  name = "Variável",
  values = c(
    temperatura_media_mensal = "blue",
    media_temperaturas_maximas_diarias = "red",
    media_temperaturas_minimas_diarias = "darkgreen"
  ),
  labels = c(
    temperatura_media_mensal = "Temperatura média mensal",
    media_temperaturas_maximas_diarias = "Média das máximas diárias",
    media_temperaturas_minimas_diarias = "Média das mínimas diárias"
  )
) +
labs(
  title = "Variação mensal da temperatura",
  x = "Mês",
  y = "Temperatura (°C)"
) +
theme_minimal() +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold"
  ),
  legend.position = "bottom"
)

```



Variável —●— Média das máximas diárias —●— Média das mínimas diárias —●— Temperatura média mensal

Criando o gráfico de precipitações

```
precipitacoes_mensais %>%
  mutate(
    mes = factor(
      mes,
      levels = 1:9,
      labels = c(
        "Jan", "Fev", "Mar", "Abr", "Mai",
        "Jun", "Jul", "Ago", "Set"
      )
    )
  ) %>%
  ggplot(aes(x = mes, y = precipitacao_mensal)) +
  geom_col(fill = "steelblue", width = 0.7) +
  labs(
    title = "Precipitação acumulada mensal - 2026",
    x = "Mês",
    y = "Precipitação acumulada (mm)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
```

